

PEAWANUCK, ON, Canada

WMO: 714340

Lat: 54.990N

Lon: 85.441W

Elev: 53

StdP: 100.69

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -37.4	(c) -35.0	(d) -41.5	(e) 0.1	(f) -37.1	(g) -39.1	(h) 0.1	(i) -34.9	(j) 9.2	(k) -16.4	(l) 8.5	(m) -16.8	(n) 2.5	(o) 250	(p) 0.810

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	27.7	17.8	25.4	16.8	23.0	16.0	19.6	24.7	18.2	23.0	16.8	21.5	5.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	12.8	21.8	16.3	11.7	20.2	14.8	10.6	19.2	56.4	24.8	51.8	22.7	47.7	21.5	30.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 9.2	(b) 8.2	(c) 7.3	(e) -40.8	(f) 32.0	(g) 2.7	(h) 2.6	(i) -42.7	(j) 33.8	(k) -44.3	(l) 35.3	(m) -45.8	(n) 36.8	(o) -47.8	(p) 38.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.3	-23.2	-21.8	-15.0	-5.9	2.8	9.9	13.8	13.4	8.6	2.1	-7.5	-17.7		
	DBStd	14.48	7.22	6.59	7.86	6.74	5.94	5.52	4.69	4.27	4.33	4.08	6.47	8.45		
	HDD10.0	5218	1029	891	775	477	241	69	13	12	77	249	525	860		
	HDD18.3	7924	1287	1124	1033	726	483	261	156	164	292	504	775	1118		
	CDD10.0	372	0	0	0	0	17	66	132	118	36	3	0	0		
	CDD18.3	36	0	0	0	0	1	7	16	11	1	0	0	0		
	CDH23.3	449	0	0	0	0	18	114	191	116	10	0	0	0		
	CDH26.7	111	0	0	0	0	2	29	47	32	1	0	0	0		
	WSAvg	3.8	3.5	3.6	3.8	4.0	4.0	3.8	3.6	3.8	3.9	4.0	3.8	3.6		
	PrecAvg	578	27	21	27	31	53	63	77	72	70	56	48	31		
	PrecMax	722	49	45	51	57	90	149	129	103	156	91	85	82		
	PrecMin	472	10	8	8	9	25	19	36	33	28	19	24	12		
	PrecStd	67	10	9	11	12	18	27	24	19	29	17	16	13		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.3	0.1	7.0	14.2	26.1	29.9	30.8	30.2	24.6	17.5	7.5	0.2		
		MCWB	-1.6	-0.5	3.3	8.4	15.1	18.7	20.1	19.4	16.9	13.9	5.5	-0.5		
	2%	DB	-5.1	-4.6	3.7	9.5	21.5	26.6	27.7	26.6	21.5	13.0	3.8	-1.5		
		MCWB	-5.4	-5.4	1.6	4.8	12.9	16.6	18.4	18.2	15.8	10.0	2.5	-2.0		
	5%	DB	-8.8	-7.8	0.8	6.5	17.2	23.6	25.4	23.6	18.6	10.2	1.8	-3.2		
		MCWB	-9.2	-8.5	-0.7	2.7	11.2	15.1	17.1	17.0	14.4	8.2	1.0	-3.6		
	10%	DB	-12.4	-11.3	-2.6	4.0	13.0	20.6	22.8	21.3	15.9	7.6	0.4	-5.2		
		MCWB	-12.8	-12.0	-4.0	1.2	8.7	14.2	16.3	15.9	12.5	6.1	-0.1	-5.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.6	-0.3	3.7	8.7	15.9	21.3	22.1	20.8	18.6	14.0	6.0	-0.3		
		MCDB	-1.2	0.3	6.2	14.1	22.8	27.4	28.1	26.9	22.8	17.1	7.0	0.2		
	2%	WB	-5.2	-5.4	1.7	5.0	13.7	18.0	19.6	19.3	16.6	10.6	2.7	-1.9		
		MCDB	-4.9	-4.7	3.5	9.0	20.6	22.8	24.6	24.7	20.2	12.5	3.6	-1.4		
	5%	WB	-9.4	-8.4	-0.7	3.0	11.4	16.1	18.1	17.9	14.8	8.3	1.1	-3.8		
		MCDB	-9.0	-7.7	0.8	6.1	17.0	22.4	23.2	22.5	18.2	10.1	1.8	-3.5		
	10%	WB	-13.0	-11.9	-4.0	1.3	8.8	14.3	16.9	16.4	12.9	6.3	0.0	-6.0		
		MCDB	-12.7	-11.2	-2.7	3.8	13.4	20.8	21.6	20.4	15.5	7.8	0.5	-5.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.3	11.3	12.4	11.0	10.2	12.8	12.4	10.6	8.6	5.8	6.6	8.0		
		MCDBR	10.4	12.3	13.3	13.1	16.7	17.0	15.4	14.4	13.0	9.7	6.0	7.1		
		MCWBR	10.6	11.5	11.5	9.2	9.4	8.7	7.8	7.7	8.0	7.2	5.4	6.8		
	5% WB	MCDBR	10.7	12.2	12.6	12.2	16.2	15.4	13.3	12.5	11.5	9.2	6.0	6.8		
		MCWBR	10.6	11.6	11.1	8.7	9.5	9.3	7.8	7.3	8.1	7.3	5.5	6.7		
(40) Clear-Sky Solar Irradiance	taub		0.233	0.253	0.288	0.369	0.378	0.374	0.395	0.372	0.343	0.303	0.273	0.221		
	taud		2.191	2.222	2.178	2.038	2.239	2.353	2.356	2.423	2.499	2.530	2.244	2.165		
	Ebn at Noon		723	843	884	844	864	872	843	843	826	786	652	648		
	Edh at Noon		57	81	109	146	127	115	113	100	81	62	55	46		
(44) All-Sky Solar Radiation	RadAvg		0.92	2.01	3.56	5.15	5.44	5.73	5.32	4.08	2.69	1.46	0.68	0.58		
	RadStd		0.08	0.14	0.25	0.34	0.52	0.66	0.46	0.35	0.28	0.13	0.10	0.09		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-41	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-41	N/A

Nomenclature: See separate page